CONSU HT-DBH Curves Using FIA Data

Basic Models (Draft 1)

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1.0 Introduction

Initial species level HT-DBH fits are presented here for species in the Continental US (CONUS) FIA data set. This is preliminary work to test models and fitting and selection methods.

2.0 Methods

The following models were selected for fitting as they are generally widely used and have good performance in published comparisons.

- model 1: HT = bh + exp(B1 + B2*DBH^(-1.0))
- model 2: HT = bh + exp(B1 + B2*DBH^B3)
- model 3: HT = bh + B1*exp(1.0-B2*DBH)^B3
- model 4: HT = bh + exp(B1+B2/(DBH+1.0))
- model 5: HT = bh + exp(B1+B2/(DBH+B3))

Models 1 and 2 are of the Schumacher (1939) type curve (also see Curtis 1967). Model 3 is the Chapman-Richards function. Models 4 and 5 are often used in FVS (Wykoff and others 1982).

Each model was fit by species that had at least 3 observations. Fits were weighted by 1/DBH (based on residual plots and published results) and were fit using the nls function in R.

3.0 Results

The data was filtered for measured height (HTCD=1) and DBH > 0 and HT > 4.5. This left 9,509,054 observations for 438 species (excluding AK). This is a lot of data and combinations, so below is a table that summarizes the fits by model (1-5 above).

- Nspecies = total number of species
- Nfit - number of species fit
- N__Bns - number of fits where B1, B23, and/or B3 were not significant (p > 0.05)
- N__Bgood - number of fits that appear to be good
- B3ns - number of fits that B3 was not different for -1 (model 2) or 1 (model 5) (P > 0.05)
- nLowRMSE - number of fits with the lowest RMSE for model number
- avgRMS - the average RMSE for all fits by model

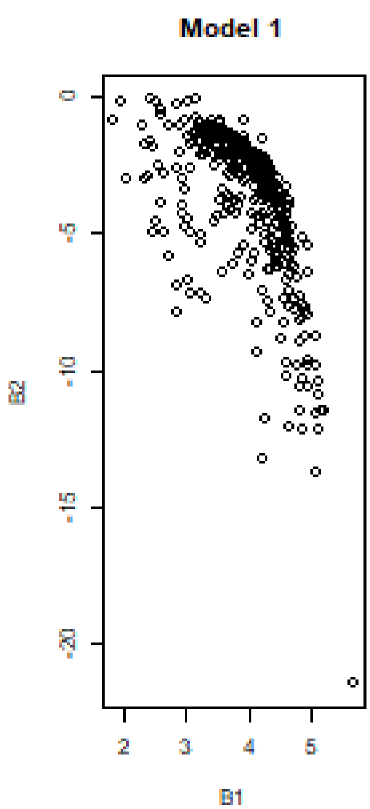
The 0.05 p-value level is arbitrary and I would probably accept higher values in this case. Also, for models 2 and 5 it may not actually be important of the B3 parameter is different form -1 or 1 respectively but is given here mostly for interest.

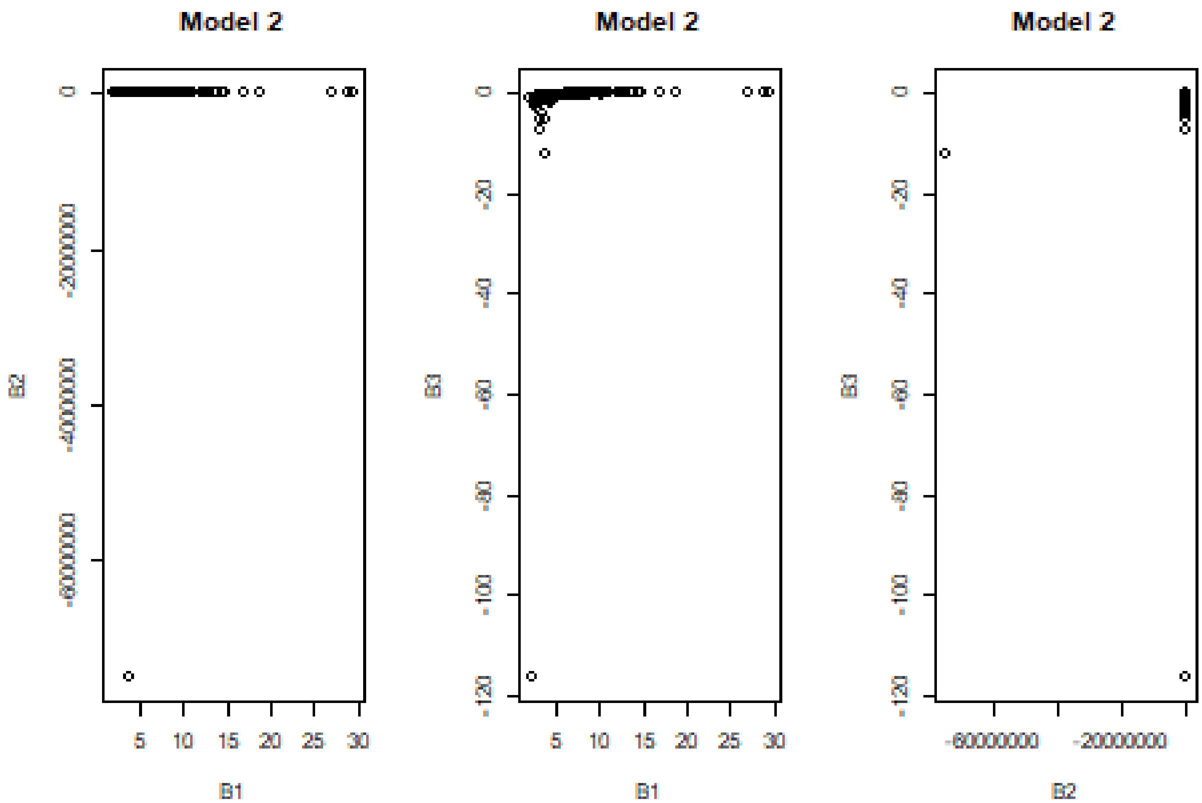
```
##
## RMSE_1 RMSE_2 RMSE_3 RMSE_4 RMSE_5
##     33     83    102     51    143

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.08858 0.22743 0.28996 0.41535 1.85302

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.3275  2.7328  3.4313  3.2268  3.8538  8.6464
```

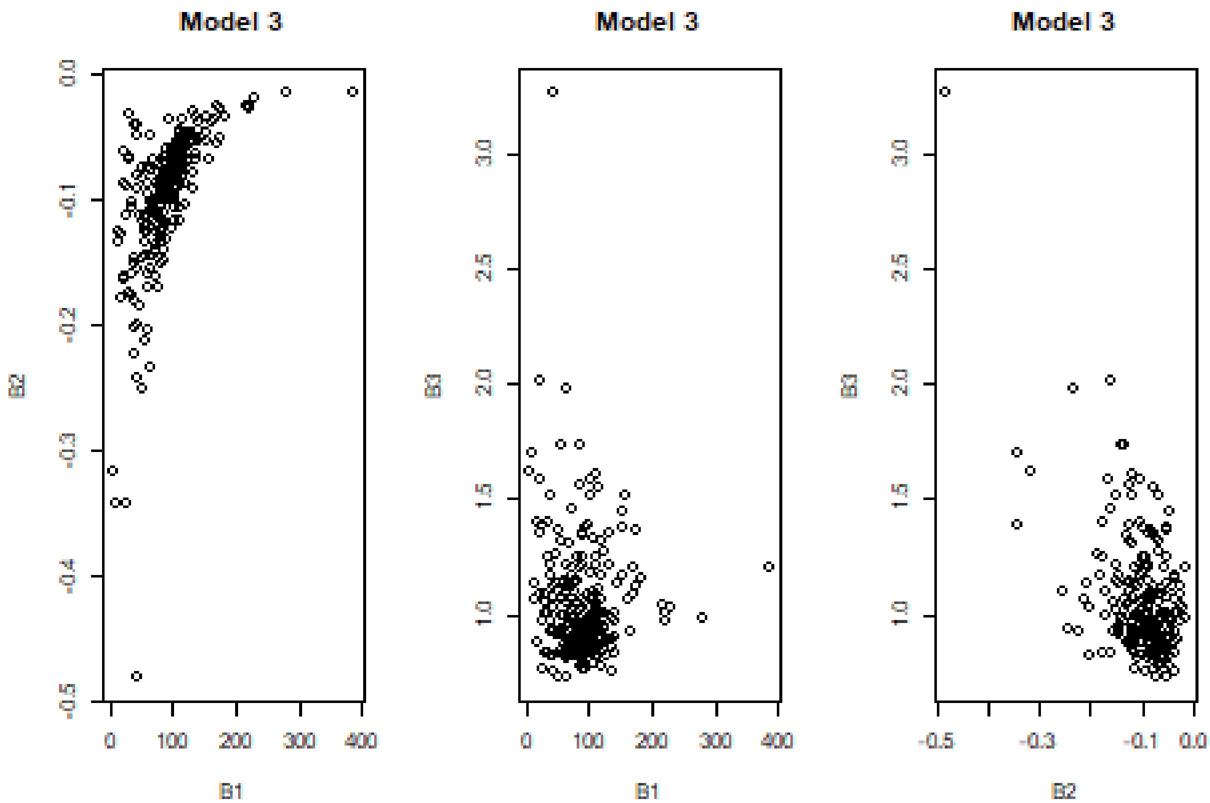
Table 1: Summary of fits by model (1-5).

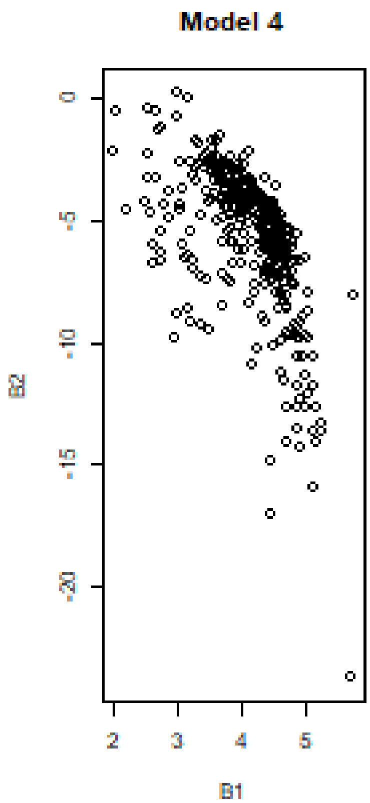
model	1	2	3	4	5
Nspecies	438	438	438	438	438
Nfit	412	338	239	412	361
N__Bns	27	48	24	26	64
N__Bgood	407	308	237	407	354
B3ns	NA	296	NA	NA	260
nLowRMSE	33	83	102	51	143
avgRMSE	3.41	3.26	3.34	3.24	3.22

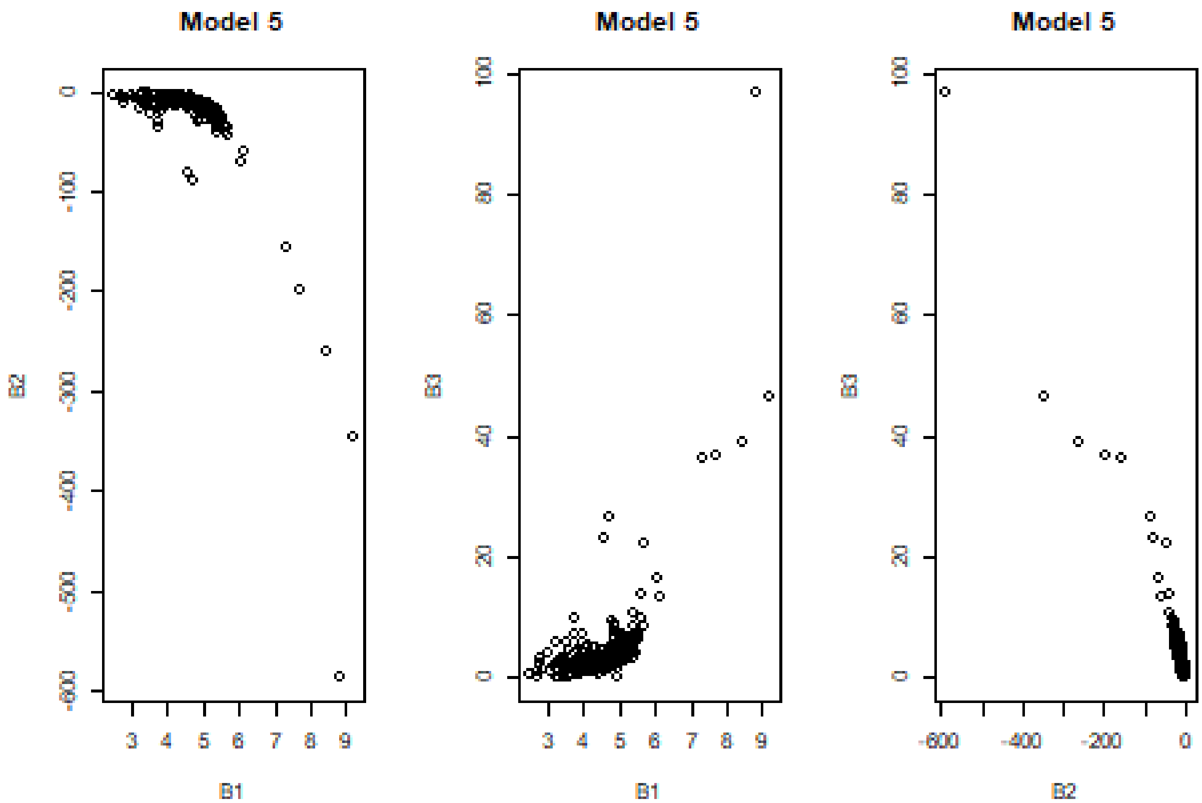
Below are plots of the coefficients. Note that there are a couple "outliers" that need to be chased down.




Model 2
Model 2
Model 2
B1
B2
B1
B3
B2
B3
5 10 15 20 25 30
0
-20000000
-40000000
-60000000
0
-20
-40
-60
-80
-100
-120
-60000000
-20000000


Model 3
Model 3
Model 3
B2
B3
B3
B1
B1
B2
0.0
-0.1
-0.2
-0.3
-0.4
-0.5
3.0
2.5
2.0
1.5
1.0
0
100
200
300
400
-0.5
-0.3
-0.1
0.0


Model 4
0
-5
-10
-15
-20
B2
2
3
4
5
B1



4.0 Discussion and Next Steps

The results here are preliminary but are a start at building and testing code for fitting a lot of HT-DBH curves.

Questions:

1. These fits were done with nonlinear least squares, however, give that plots have remeasurements (within plot correlation structure), should we use mixed models (MM) with random effects for plots?

Statistically thinking the answer is probably yes (especially if someone wants to publish this mess). However, I am not sure of the overall advantage beyond possibly making some reviewer happy.

Fitting regional HD-DBH curves is a bit of a problem because they are very likey biased for individual stands/plots which tend to have much flatter curves than the regional fits (using many stands/plots). This means users are going to have to do calibrations or live with the bias (unknown impacts on volume and growth) – unfortunately I doubt many do. Using the fixed effects of a MM fit will not fix this. Both MM or least squares (LS) models will require localizing (calibrating) fits with new stand/plot data. For MM this is done by estimating the random effect with variance/covariance structures. This requires derivatives and some fun matrix algebra. I don't see the average user doing this for each new run and so an application would likely need to be built in for processing (Hann has a calibration function in ORGANON which uses a LS approach). Alternatively, calibrating the models use am OLS (predicted to actual) approach is much easier to do and should work for either LS or MM fits. Temesgen and others (2008) look at different calibration approaches and (my opinion) showed little differences (maybe a few less trees need for the MM approach). The testing I did seems to confirm this (FIA_HTDBH_modeling_20260530.pdf).

2. There appears to be a few “outliers” measurements that should be addressed (will take some time to run through all the species).

3. Are there other models that should be considered?

4. I assume we probably want to use just a single model form. What should be the criteria? Fortunately the differences in RMSE are typically small between the model fits. I currently don’t have it output but I probably should add something like the mean difference to the fitting output.

5. If we used the LS approach (rather than MM) I used “nls” in R but should try “nlsLM” which tends to be a bit more robust and might (maybe) converge on a few more species.

5.0 References

Curtis, R.O. 1967. Height-diameter and height-diameter-age equations for second-growth Douglas-fir. Forest Science 14(4):365-375.

Schumacher, F. X. (1939). A New Growth Curve and Its Applications to Timber Yield Studies. Journal of Forestry, 37, 819-820. [Referenced by others]

Temesgen, H.; Monleon, V.J.; Hann, D.W. 2008. Analysis and comparison of nonlinear tree height prediction strategies for Douglas-fir forest. Canadian Journal of Forest Research 38(3): 553-565.

Wykoff, William R. 1986. Supplement to the user’s guide for the stand prognosis model-version 5.0. Gen. Tech. Report INT-208. Ogden, UT: U.S. Department of Agriculture, Intermountain Forest and Range Experiment Station. 36 p.

Wykoff, William R.; Crookston, Nicholas L.; Stage, Albert R. 1982. User’s Guide to the stand prognosis model. Gen. Tech. Report INT-133. Ogden, UT: U.S. Department of Agriculture, Intermountain Forest and Range Experiment Station. 112 p